



BRAC UNIVERSITY

CSE461: Introduction to Robotics

Lab Project Report

Title: *Autonomous Multi Point Water Quality Survey Robot*
by

[Group_01]

[Section_05]

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Abstract

Our project is an autonomous water quality testing robot. It floats on water, detects the station, goes there and tests the quality of water. To test the quality of water, we are considering parameters like pH, turbidity and temperature. Our robot steps down the 11.1 V from battery to 5 V for the arduino while motor drivers directly get power from the battery to run the motor. It uses sensors like pH sensor, turbidity sensor and temperature sensor to measure the parameters and test the water. It can detect pollution and warn us. In a broader aspect it can be improved with more sensors and features and work as an environmental asset to analyze water conditions enabling remote monitoring.

Keywords

Arduino Mega 2560, Water Quality testing, Sensing, Autonomous, Floating body

1. Introduction

Water quality is a huge concern in our country as well as in the world. Water that looks clean can also be chemically unusable. While many projects exist which measure water quality, there is still a major gap in those as most of them require manual sample collection, labour and limited testing place.

Our water quality testing robot is built as an autonomous robot which can float through station to station and test water itself, collecting and recording the measurements for judgement. It requires less labor or manual work. It is related to computer interfacing as it connects sensors like pH sensor, turbidity sensor and temperature sensor to microcontroller. The microcontroller we used is an Arduino Mega 2560.

2. Related Work/ Inspiration

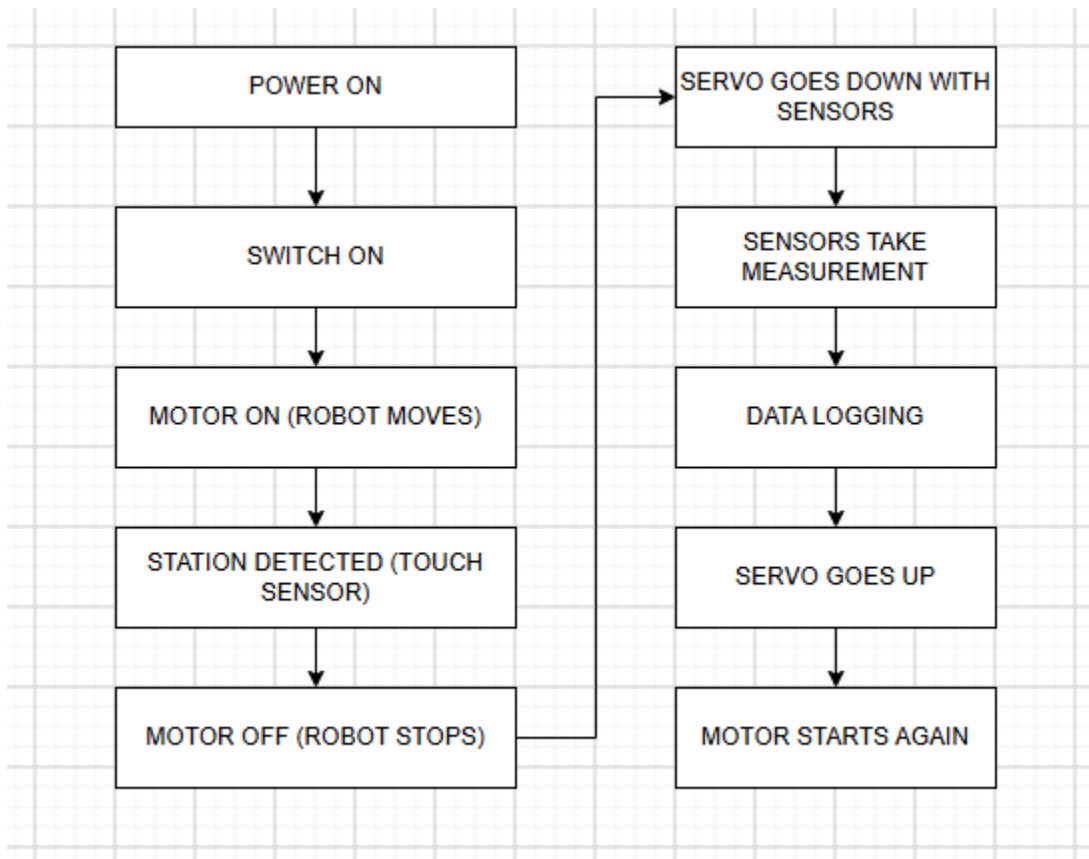
We saw similar projects which test water quality. But it was limited to measuring one or two parameters and all of them were stationary. People had to bring experimental substances (water from different places) to it. Moving robots were remote controlled which required a person's presence all the time to control it and it was expensive as well.

We designed the robot in a way that it addresses all the existing shortcomings. The biggest takeaway is that this robot is autonomous, it moves on its own, records the data. So full time monitoring is not required. It moves to different places in a floating region, so we can easily test water of a big water flow instead of bringing water from different parts of the same water reservoir. We tried to make it budget friendly as well.

3. Technical Approach

System Architecture:

The system measures quality of water by detecting stations and only does its testing there. It uses a power supply, a switch that turns on after getting power. It uses a motor driver to drive the motors and a touch sensor module to detect stations. Sensors (pH sensor, turbidity sensor, temperature sensor) to test the quality of water and saves the data in SD card via a SD card module



Here is the Component list & Price breakdown

Component	Price (bdt)
pH electrode sensor probe and monitoring module kit	2550
Waterproof DS18B20 Compatible Digital Temperature Sensor	220

Water Turbidity Sensor Detection Module Kit for Arduino	830
TTP223 Touch Sensor Module	50
Hobby Gearmotor 6V TT Dual Shaft BO Motor	2 x 95 = 190
BLX-A Cover with Fuse Holder	50
Fuse 5A	10
MicroSD Card Module for Arduino	75
Hobbywing 3A UBEC 5V	670
16A 250V Double Rocker Switch SPST ON-OFF 6 Terminals with Waterproof Cover	75
Arduino mega 2560 r3	600 (rent price)
Motor driver bts7960 x 2	376 (rent price)
Servo mg996r	140 (rent price)
11.7V battery with holder	140 (rent price)
Battery charger	100 (rent price)
Breadboard (large)	64 (rent price)
Breadboard (small)	55
Jumper wire (around 100)	250
Resistor	15

Circuit/Schematic:

Functionality: The system gets power from the battery, when the power gets on, the switch is on. Then the robot starts. It detects the sensor through its touch sensor module. When the touch sensor detects a station, its motors are stopped meaning it stops there and the servo goes down with the sensors. The sensor takes the measurements and it is saved in the SD card via SDcard module. After the data is taken the motor starts again. In this way it can detect and go to multiple stations and test the water of that particular station.

Workflow (From turning on to detecting)

POWER ON -> SWITCH ON MOTOR ON (ROBOT MOVES) -> STATION DETECTED (TOUCH SENSOR) -> MOTOR OFF (ROBOT STOPS) -> SERVO GOES DOWN WITH SENSORS -> SENSORS TAKE MEASUREMENT -> DATA LOGGING -> SERVO GOES UP -> MOTOR STARTS AGAIN

Challenges: We faced several **challenges** throughout the project. The first challenge was to choose the correct and co-related components. We had to search, study, watch youtube videos to finally come to our final list. When we built the circuit, at first there were some wiring issues. The temperature sensor kept showing -127 degrees. Later we checked and realised it was not the component that failed but it is a function that the temperature sensor shows this value if there is a loose wire. We undo all the circuit and build block by block while testing all the components. While building the chassis we realized, the tt motor is mainly for wheels not for boat propellers, we had to diy make the propeller. Building the chassis was also challenging keeping enough space and maintaining weight so that it floats and runs. We watched several youtube videos to understand it.

4. Sustainability & Impact

The water quality testing robot built here is affordable, sustainable and environmental friendly. It successfully measures parameters like pH, turbidity and temperature within low cost and gives a reusable platform. It can detect water pollution. Its functionality reduces manual work. It integrates sensors, actuators to successfully do robotic movement. While it only works in limited/ fixed stations and indoor areas, with further improvements, it can work in a broader range and in large water reservoirs. It can be furtherly be improved by adding sensors like tds sensor, water leak detection sensor to detect water condition more accurately and safely. We can add wireless connection, gps tracking, bluetooth controlling etc making it a more efficient water quality testing robot that can test water quality in remote areas on its own and analysing data as well

5. Results & Discussion

Test sample	pH	Turbidity	Temperature	Observation
Tap water	7.1	Low	21 degree	Clean, neutral pH range
Bottle water	7.25	Low	28 degree	Clean, pH slightly changes despite being from the same source, still in neutral pH range
Pond	6.65	High	30 degree	Visibly Cloudy, slightly acidic
Pond	-	-	-127 degree	Wiring Issue

From our tested several results, we see the sensors show accurate results of the water. However, the pH sensor is slightly sensitive to temperature, so the same water reads different slightly pH values when temperature changes. The temperature sensor assures that the water did not suddenly have chemical change, it is the effect of changing temperature on the pH sensor. Also the temperature sensor shows -127 degree while the other sensors show no result whenever there is a wiring issue. So the temperature sensor also works as an identifier of sensor connection or system falls during testing/working.

6. Conclusion

This project gives us a sustainable, environment friendly robot that can measure the quality of water and detect water pollution. Water pollution is a world wide problem. This project can assure us which water is safe. It is autonomous and reduces manual water testing in several stations by doing the work on its own. It integrates a microcontroller, sensors and actuator to integrate a movable robot. With further development it can collect , store and analyze water sampling data more efficiently and can be expanded to larger levels.

7. Contribution

Name	ID	Role(s) / Responsibilities	Specific Contributions
NISHAT TASNIM ZANNAT	23201116	Project Management & Documentation	● Overviewed the reports ● Chose components, made the final component list and managed the components ● Assisted in circuit + chassis building

FATIMA TUZ JOHORA	21301625	Circuit Design & Implementation:	● Made the initial circuit design ● Did the initial wiring ● Assisted in circuit + chassis building
MD. TASHFIQ BIN SARWAR	22201607	Hardware Integration & Troubleshooting	● Identified the components and resolved existed issues ● Tested the circuit block by block and made the final design ● Assisted in circuit + chassis building
MALIHA NIGAR	22299012	Hardware Handling	● Handled logistic and hardware side ● Stored the circuit ● Assisted in building circuit +chassis
RAYA TASNIM	23101038	X	X

References

<https://youtu.be/vOCkCpksv04?si=-QsmfYUGMEPWu216>

https://youtu.be/He_BNxd0vWw?si=BL703AcbJHjSpDQk